

Spendly

Offline-First · Zero-Knowledge · Peer-to-Peer Finance Ecosystem

Semester VII · Information Technology

Student Name · Enrollment Number · College Name



The Problem Is Everywhere

Modern finance systems are fragile by design — exposing users to privacy breaches, downtime, and dependency.

Privacy at Risk

Centralized databases create single points of failure — one breach exposes everything.

Always Online

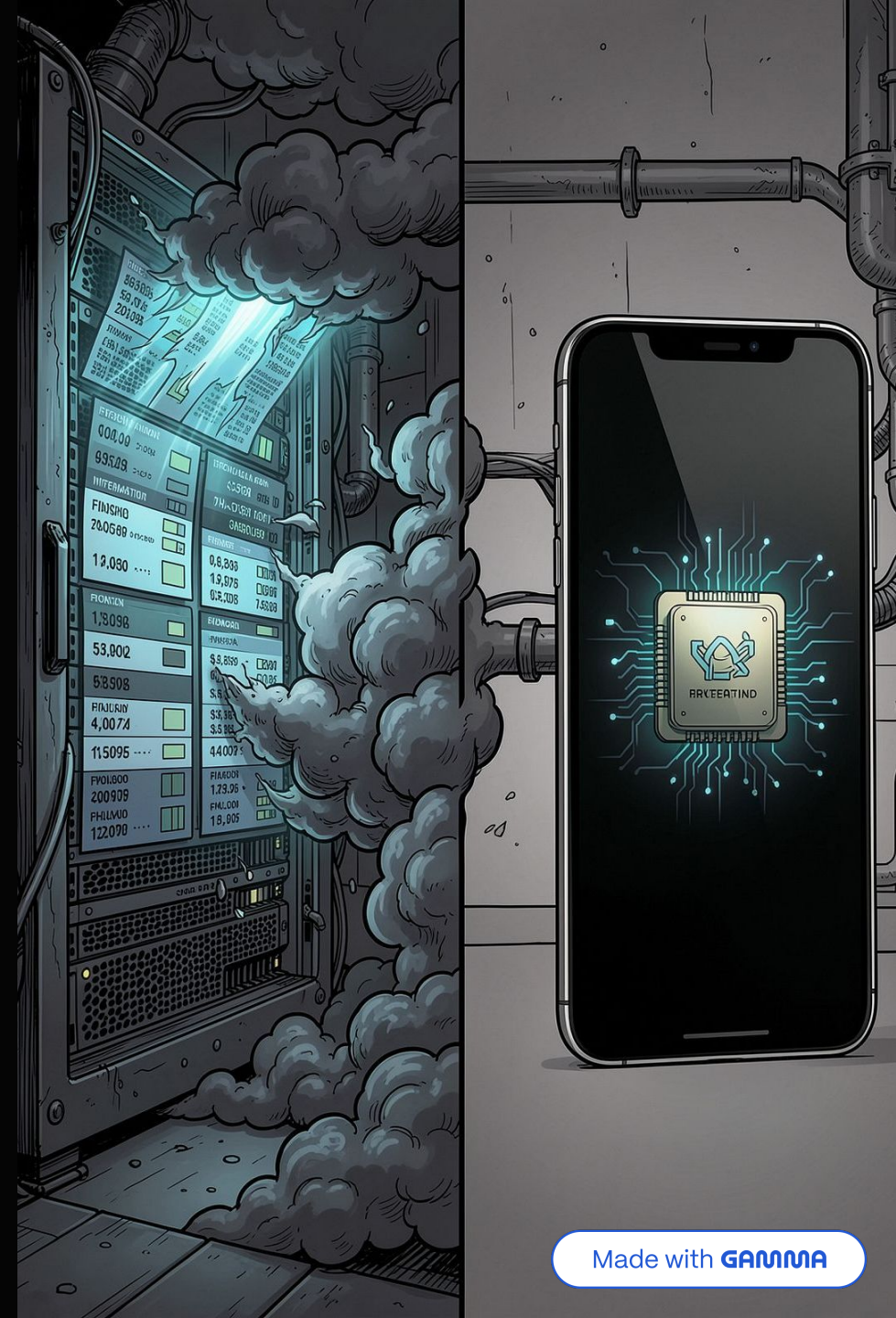
Cloud dependency means no service in low-connectivity or offline environments.

Recurring Costs

Monthly POS subscriptions burden small merchants with unnecessary overhead.

Data Breaches

Centralized storage is a permanent target — financial data is never truly safe.





Introducing Spendly

Three applications. One ecosystem. Zero compromise.

Spendly

The consumer app. Offline-first expense tracking, encrypted local storage, and peer-to-peer merchant communication — no internet required.

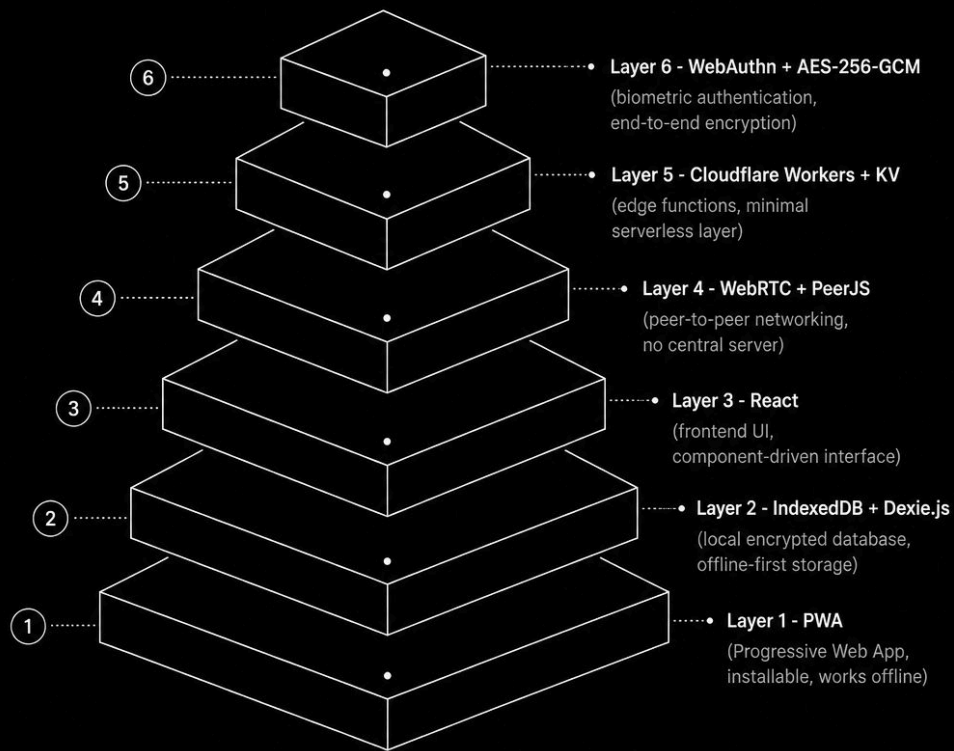
Spendly Shop

The merchant terminal. Inventory management, QR invoicing, cash wallet, and EMI alerts — all running locally on-device.

Spendly Radar

The discovery layer. Real-time offer detection, coupon validation, and proximity-based deal sharing between peers.

System Architecture



Every Layer, Purpose-Built

Spendly's architecture is an exploded stack of transparent, independent layers — each chosen for resilience, privacy, and performance.

- **React** — Component-driven, responsive UI layer
- **IndexedDB + Dexie.js** — Encrypted offline-first local database
- **WebRTC + PeerJS** — Direct device-to-device communication
- **Cloudflare Workers + KV** — Minimal edge serverless functions
- **WebAuthn + AES-256-GCM** — Hardware-backed biometric security
- **PWA** — Installable, works without internet

How It Works

01

Offer Discovery

Spendly Radar detects nearby merchant offers via peer-to-peer proximity.

02

Coupon Claim & Barcode Scan

User claims offer; barcode scanned for product lookup and validation.

03

Offer Validation

Zero-knowledge proof verifies eligibility without exposing user data.

04

Invoice Generation & Sharing

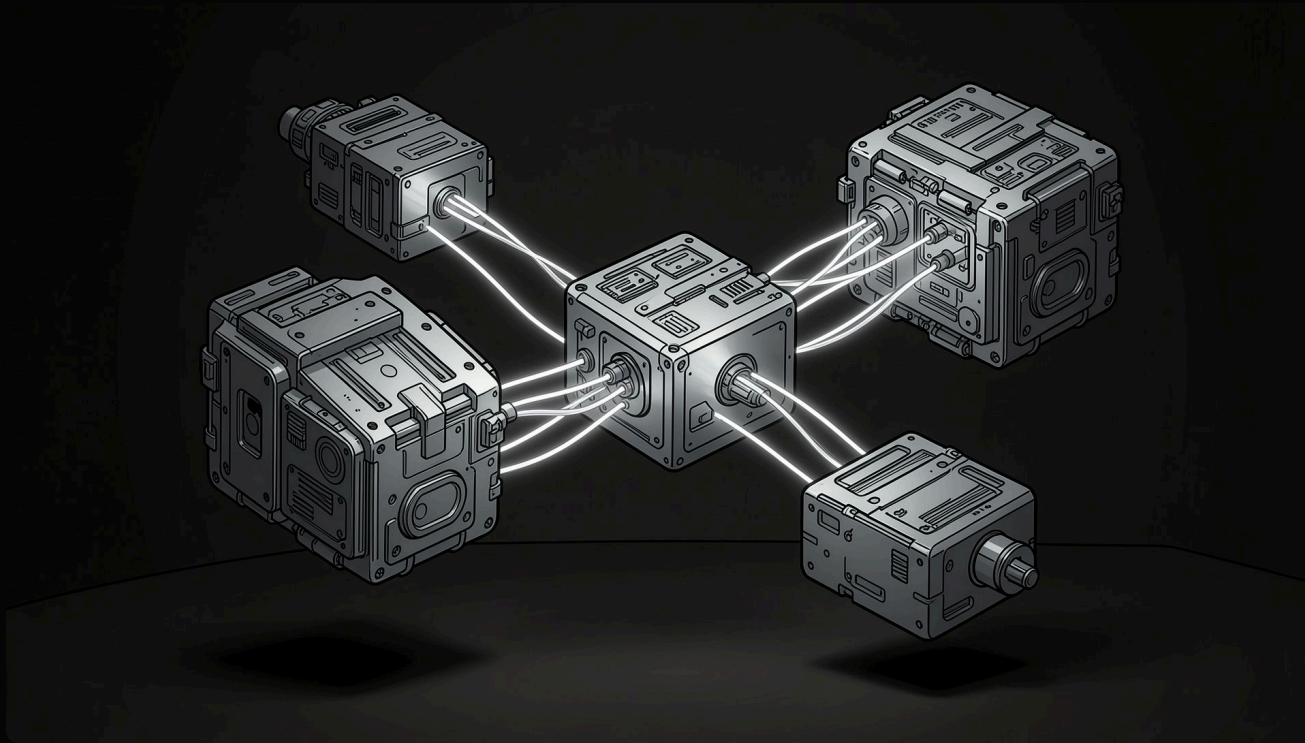
Encrypted invoice created locally and shared peer-to-peer instantly.

05

Expense Saving & Analytics

Transaction logged offline; analytics generated on-device with full privacy.

Technology Stack



Frontend

React · PWA · Responsive UI

Backend & Edge

Cloudflare Workers · KV Storage

Networking

WebRTC · PeerJS · Offline Sync

Database

IndexedDB · Dexie.js

Security

WebAuthn · AES-256-GCM · Secure Enclave

Deployment

Installable PWA · Zero Server Infrastructure

Security by Design

Spendly's security model is not a feature — it is the foundation. Every transaction, every record, every authentication event is designed to never leave the device unless explicitly intended.

Zero-Knowledge

No server ever sees your financial data. Proofs are verified without revealing underlying information.

Offline-First

All data resides in IndexedDB — encrypted, local, and inaccessible to external threats.

WebAuthn + Biometric

Hardware-backed authentication via fingerprint or face — no passwords, no replay attacks.

AES-256-GCM

Military-grade encryption applied to every stored record and peer-to-peer transmission.



Features



Expense Tracking

Real-time categorization and spending insights — all processed on-device.



Merchant Billing

Instant QR invoice generation with offline capability and peer delivery.



Inventory & Cash Wallet

Stock management and digital cash ledger for merchants — no internet needed.



EMI Alerts & Analytics

Automated payment reminders and on-device financial analytics with full privacy.



Advantages & Future Scope

Core Advantages

- **Offline-First** — Fully functional without internet connectivity
- **Zero Server Cost** — No infrastructure, no subscriptions, no overhead
- **Privacy-First** — Zero-knowledge architecture protects every user
- **Fast Billing** — Instant peer-to-peer invoice exchange via WebRTC
- **P2P Communication** — Direct device-to-device, no intermediaries

Future Scope

- **AI Insights** — On-device spending pattern intelligence
- **Voice Billing** — Natural language transaction commands
- **UPI & NFC** — Native payment rail integration
- **GST Compliance** — Automated tax invoice generation
- **Multi-Store** — Merchant network management at scale

Spendly

A finance ecosystem built for the world as it is — not as we wish it to be.
Offline. Private. Peer-to-peer.

Thank You · Questions Welcome

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